

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF APPLIED MATHEMATICS
AMA3640 STATISTICAL INFERENCE

Assignment 2

Due at 11:59pm on 9 November 2025

1. Let $X_1, \dots, X_n$ be a random sample from the Bernoulli distribution with mean p . Suppose that $n \geq 3$. Show that $\hat{\theta} \equiv X_1 X_2 (1 - X_3)$ is an unbiased estimator of $\theta \equiv p^2(1 - p)$. Use the Rao–Blackwell theorem, $\hat{\theta}$, and the sufficient statistic $\sum_{i=1}^n X_i$ to find a better unbiased estimator of θ .
2. Consider a random sample $X_1, \dots, X_n$ from the Bernoulli distribution with success probability θ .

(a) Show that $S^2 \equiv \sum_{i=1}^n (X_i - \bar{X})^2 / (n - 1)$ is unbiased for $\theta(1 - \theta)$, where $\bar{X} = \sum_{i=1}^n X_i / n$.

(b) We know that

$$\begin{aligned} \text{Var}(S^2) = \frac{1}{n(n-1)^2} & \left(n^2\theta - 5n^2\theta^2 + 12n\theta^2 - 2n\theta + 8n^2\theta^3 - 20n\theta^3 \right. \\ & \left. + \theta - 7\theta^2 + 12\theta^3 - 6\theta^4 + 10n\theta^4 - 4n^2\theta^4 \right). \end{aligned}$$

Does S^2 attain the Cramér–Rao lower bound?

(c) Argue that $\sum_{i=1}^n X_i$ is a complete sufficient statistic for θ .

(d) Is S^2 the best unbiased estimator of $\theta(1 - \theta)$? Justify your claim.

3. Let X_1 and X_2 be two independent and identically distributed variables with probability mass function

$$f(x; \theta) = \frac{1}{2^{x-\theta+1}} \quad \text{for } x = \theta, \theta + 1, \dots$$

where $\theta \in \{0, 1, 2, \dots\}$.

(a) Show that $X_{(1)} \equiv \min(X_1, X_2)$ is a sufficient statistic for θ .

(b) Find $E(X_{(1)})$.

(c) Find the best unbiased estimator of θ . You can use the fact that $X_{(1)}$ is a complete statistic for θ .

4. Let X_1 and X_2 be independent $\text{Uniform}(\theta, \theta + 1)$ random variables. Consider testing $H_0 : \theta = 0$ versus $H_1 : \theta = 0.5$. We have two tests:

T_1 : Reject H_0 if $X_1 > 0.95$;

T_2 : Reject H_0 if $X_1 + X_2 > C$.

- (a) Find the value of C such that T_2 has the same size as T_1 . Hint: Under H_0 ,

$$P(X_1 + X_2 > C) = \int_0^1 P(X_2 > C - x) dx.$$

- (b) Evaluate the power of the two tests, where for T_2 , use the value of C you obtained in (a).

5. Suppose that X is a continuous random variable with density

$$f(x; \theta) = 1 - \theta^2 \left(x - \frac{1}{2} \right) \quad \text{for } 0 < x < 1,$$

where $-1 < \theta < 1$.

- (a) Find the most powerful size α test for $H_0 : \theta = 0$ versus $H_1 : \theta = 0.5$.
(b) Argue that the test in (a) is the UMP size α test for $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$.

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Assignment 2 Solution

1. Because $X_1, \dots, X_n$ are independent,

$$E(X_1 X_2 (1 - X_3)) = E(X_1)E(X_2)E(1 - X_3) = p^2(1 - p).$$

Therefore, $\hat{\theta}$ is unbiased for $\theta = p^2(1 - p)$.

By the Rao–Blackwell theorem, an improved unbiased estimator is $E(\hat{\theta} \mid \sum_{i=1}^n X_i)$. Let $T = \sum_{i=1}^n X_i$. We compute the conditional expectation as follows.

For $t < 2$, it is not possible for both X_1 and X_2 to be 1, so

$$E(X_1 X_2 (1 - X_3) \mid T = t) = 0.$$

Likewise, the conditional expectation is 0 if $t = n$. For $2 \leq t \leq n - 1$, note that $X_1 X_2 (1 - X_3) = 1$ if and only if $X_1 = 1$, $X_2 = 1$, and $X_3 = 0$. This corresponds to assigning 2 of the t ones to X_1 and X_2 , and the remaining $t - 2$ ones among the $n - 3$ other variables (excluding X_3). Therefore,

$$P(X_1 = 1, X_2 = 1, X_3 = 0, \sum_{i=1}^n X_i = t) = \binom{n-3}{t-2} p^t (1-p)^{n-t}$$

and

$$P(\sum_{i=1}^n X_i = t) = \binom{n}{t} p^t (1-p)^{n-t}.$$

Therefore,

$$\begin{aligned} E(X_1 X_2 (1 - X_3) \mid T = t) &= \frac{P(X_1 = 1, X_2 = 1, X_3 = 0, \sum_{i=1}^n X_i = t)}{P(\sum_{i=1}^n X_i = t)} \\ &= \frac{\binom{n-3}{t-2}}{\binom{n}{t}} \end{aligned}$$

for $t = 2, \dots, n - 1$, and 0 otherwise.

Therefore, the Rao–Blackwellized estimator is

$$\tilde{\theta} = \begin{cases} \frac{\binom{n-3}{t-2}}{\binom{n}{t}} & \text{if } 2 \leq T \leq n - 1, \\ 0 & \text{otherwise.} \end{cases}$$

This estimator is unbiased and has lower variance than $\hat{\theta}$.

2. (a) Note that

$$\begin{aligned}
E\left\{\sum_{i=1}^n (X_i - \bar{X})^2\right\} &= E\left\{\sum_{i=1}^n X_i^2 - \frac{1}{n}\left(\sum_{i=1}^n X_i\right)^2\right\} \\
&= \sum_{i=1}^n E(X_i^2) - \frac{1}{n}E\left\{\left(\sum_{i=1}^n X_i\right)^2\right\} \\
&= n\theta - \frac{1}{n}[n\theta(1-\theta) + (n\theta)^2] \\
&= n\theta - \theta(1-\theta) - n\theta^2 \\
&= (n-1)\theta(1-\theta).
\end{aligned}$$

Therefore, $E(S^2) = \theta(1-\theta)$.

- (b) The Fisher information for θ is

$$I(\theta) = \frac{1}{\theta(1-\theta)}.$$

The Cramér–Rao lower bound for unbiased estimators of $\theta(1-\theta)$ is

$$\frac{(1-2\theta)^2\theta(1-\theta)}{n}.$$

Since $\text{Var}(S^2)$ does not attain this bound (as can be checked by comparing the given variance formula to the CRLB), S^2 does not attain the Cramér–Rao lower bound.

- (c) The Bernoulli pmf can be written as

$$f(x; \theta) = \theta^x(1-\theta)^{1-x} = (1-\theta)e^{x \log\{\theta/(1-\theta)\}},$$

which is an exponential family with natural sufficient statistic $\sum_{i=1}^n X_i$. The set $\{\log(\theta/(1-\theta)) : 0 < \theta < 1\}$ is an open interval, so $\sum_{i=1}^n X_i$ is complete and sufficient for θ .

- (d) S^2 is a function of the complete sufficient statistic $\sum_{i=1}^n X_i$ and is unbiased for $\theta(1-\theta)$. By the Lehmann–Scheffé theorem, S^2 is the best unbiased estimator of $\theta(1-\theta)$.

3. (a) The joint probability mass function is

$$\frac{I(x_1 \geq \theta)}{2^{x_1-\theta+1}} \frac{I(x_2 \geq \theta)}{2^{x_2-\theta+1}} = \frac{I(\min(x_1, x_2) \geq \theta)}{2^{x_1+x_2-2\theta+2}}.$$

By the factorization theorem and setting $h(\mathbf{x}) = (2^{x_1+x_2+2})^{-1}$, $X_{(1)}$ is a sufficient statistic for θ .

(b)

$$\begin{aligned}
E(X_{(1)}) &= \sum_{k=\theta}^{\infty} k \frac{3}{2^{2k-2\theta+2}} \\
&= \sum_{k=0}^{\infty} (k + \theta) \frac{3}{2^{2(k+\theta)-2\theta+2}} \\
&= \sum_{k=0}^{\infty} k \frac{3}{2^{2k+2}} + \sum_{k=0}^{\infty} \theta \frac{3}{2^{2k+2}} \\
&= \sum_{k=0}^{\infty} k \frac{3}{4} \left(\frac{1}{4}\right)^k + \theta \\
&= \frac{1}{3} + \theta.
\end{aligned}$$

(c) Since $E(X_{(1)} - 1/3) = \theta$ and $X_{(1)} - 1/3$ is a function of a complete sufficient statistic, $X_{(1)} - 1/3$ is the best unbiased estimator for θ .

4. (a) For T_1 , the size is

$$P(X_1 > 0.95) = 1 - 0.95 = 0.05.$$

For T_2 , we want to find C such that $P(X_1 + X_2 > C) = 0.05$ under H_0 . Consider $0 < C \leq 1$. For $0 \leq x \leq C$,

$$P(X_2 > C - x) = 1 - (C - x) = x - C + 1.$$

For $C \leq x \leq 1$, $C - x \leq 0$, so $P(X_2 > C - x) = 1$. Therefore,

$$\begin{aligned}
P(X_1 + X_2 > C) &= \int_0^C (x - C + 1) dx + \int_C^1 1 dx \\
&= \left[\frac{1}{2}x^2 - Cx + x \right]_0^C + (1 - C) \\
&= \left(\frac{1}{2}C^2 - C^2 + C \right) + (1 - C) \\
&= -\frac{1}{2}C^2 + C + 1 - C \\
&= 1 - \frac{1}{2}C^2.
\end{aligned}$$

Alternatively, consider $1 < C \leq 2$. For $0 \leq x \leq C - 1$, $C - x > 1$, so $P(X_2 > C - x) = 0$. For $C - 1 \leq x \leq 1$,

$$P(X_2 > C - x) = 1 - (C - x) = x - C + 1.$$

Therefore,

$$\begin{aligned}
P(X_1 + X_2 > C) &= \int_0^{C-1} 0 \, dx + \int_{C-1}^1 (x - C + 1) \, dx \\
&= \int_{C-1}^1 (x - C + 1) \, dx \\
&= \left[\frac{1}{2}x^2 - Cx + x \right]_{x=C-1}^{x=1} \\
&= \frac{1}{2}C^2 - 2C + 2.
\end{aligned}$$

Therefore,

$$P(X_1 + X_2 > C) = \begin{cases} 1 - \frac{1}{2}C^2, & 0 < C \leq 1 \\ \frac{1}{2}C^2 - 2C + 2, & 1 < C \leq 2 \end{cases}$$

To match the size of T_1 , set $P(X_1 + X_2 > C) = 0.05$ and solve for C in the $1 < C \leq 2$ case:

$$\frac{1}{2}C^2 - 2C + 2 = 0.05 \implies C^2 - 4C + 3.9 = 0$$

$$C = \frac{4 \pm \sqrt{16 - 4 \times 3.9}}{2} = \frac{4 \pm \sqrt{0.4}}{2}$$

Since $1 < C \leq 2$, take the smaller root:

$$C = \frac{4 - 0.6325}{2} = 1.6838$$

(b) For T_1 , the power is

$$P(X_1 > 0.95; H_1) = \frac{1.5 - 0.95}{1.5 - 0.5} = 0.55.$$

For T_2 , the sum $X_1 + X_2$ has support $[1, 3]$. The power is

$$P(X_1 + X_2 > C) = \int_{0.5}^{1.5} P(X_2 > C - x) \, dx$$

where $C = 1.6838$. Note that $X_1 - 0.5 \sim \text{Uniform}(0, 1)$, and the same for X_2 , so $P(X_1 + X_2 - 1 > C; H_1) = P(X_1 + X_2 > C; H_0)$. Therefore, under H_1 ,

$$P(X_1 + X_2 > 1.6838) = P(X_1 + X_2 - 1 > 0.6838) = 1 - \frac{1}{2}(0.6838)^2 = 0.7662.$$

The power of T_2 is 0.7662.

5. (a) By the Neyman–Pearson lemma, the most powerful test takes the form of

$$\text{reject } H_0 \text{ if and only if } \frac{f(X; 0.5)}{f(X; 0)} > k$$

for some k . Note that

$$\frac{f(X; 0.5)}{f(X; 0)} = 1.125 - 0.25x,$$

so the most powerful test rejects H_0 if and only if $X < k'$ for some k' . Since $X \sim \text{Uniform}(0, 1)$ under H_0 , we set $k' = \alpha$ for a size α test. The most powerful size α test is to reject H_0 if and only if $X < \alpha$.

- (b) For testing $H_0 : \theta = 0$ versus $H_1 : \theta = \theta_1 \neq 0$, the rejection region of the most powerful test takes the form of

$$1 - \theta_1^2 \left(X - \frac{1}{2} \right) > k \iff X < \frac{1 - k}{\theta_1^2} + \frac{1}{2}.$$

The most powerful size α test is to reject H_0 if and only if $X < \alpha$ regardless of the value of θ_1 . Therefore, the test in (a) is the UMP size α test for $H_0 : \theta = 0$ versus $H_1 : \theta \neq 0$.